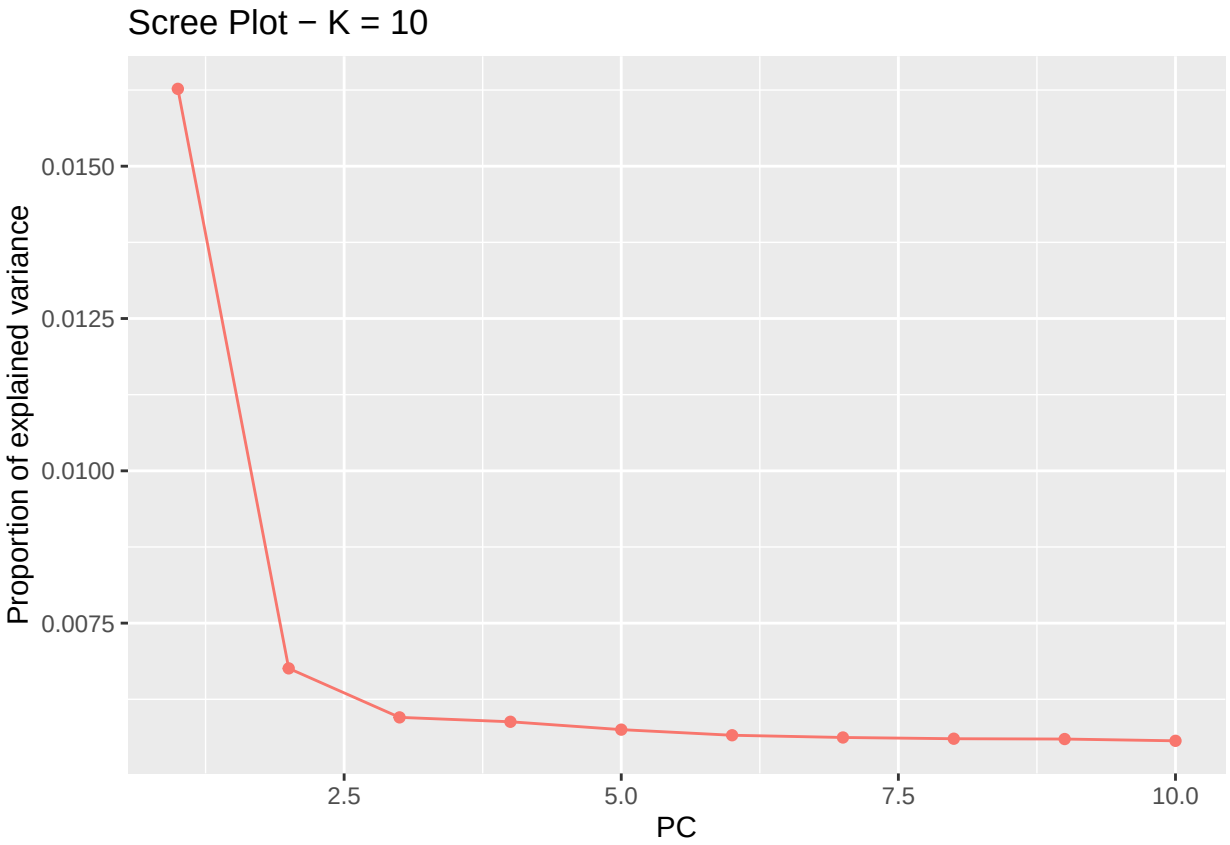
PCA

2024-01-31

PCA's using Malagasy only data

First, choosing number of PC's (K) from scree plot of percentage variance explained.

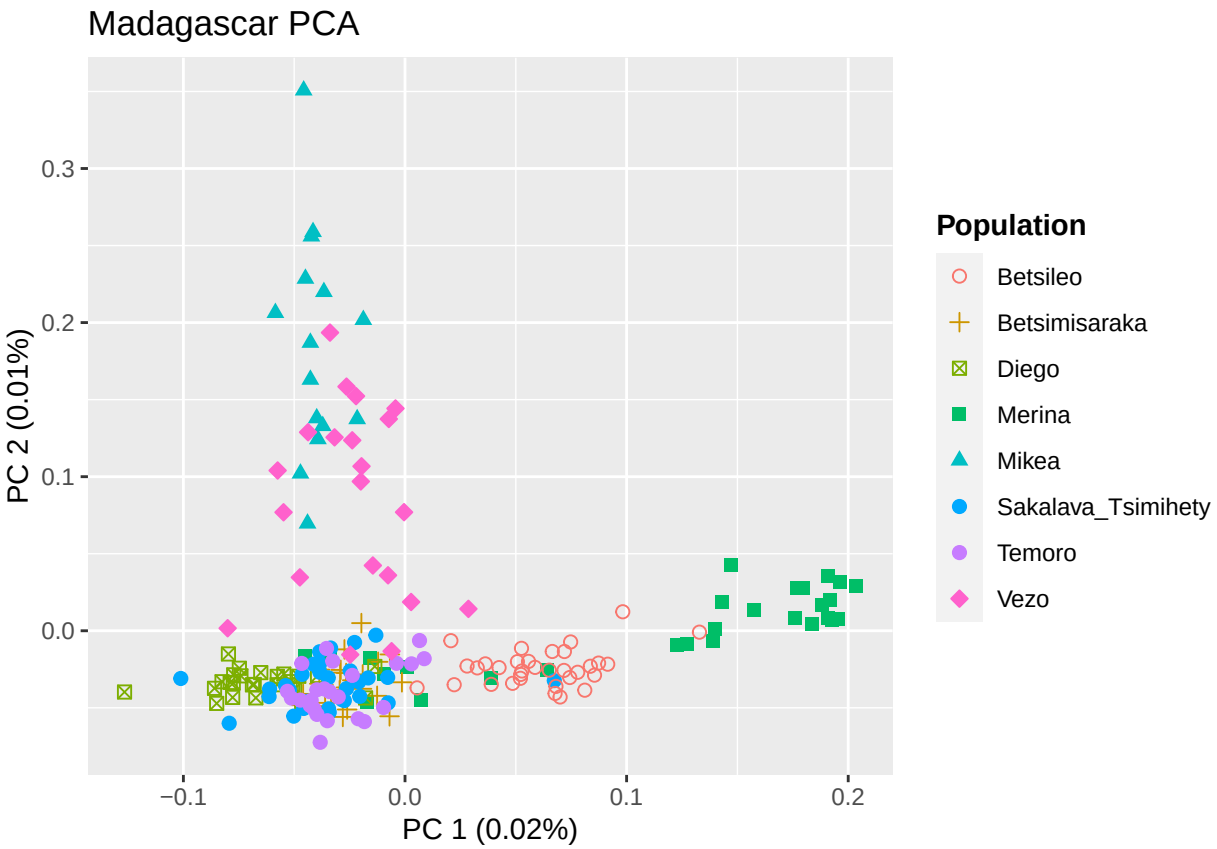


Sharp drop around 2, K is set to 3.

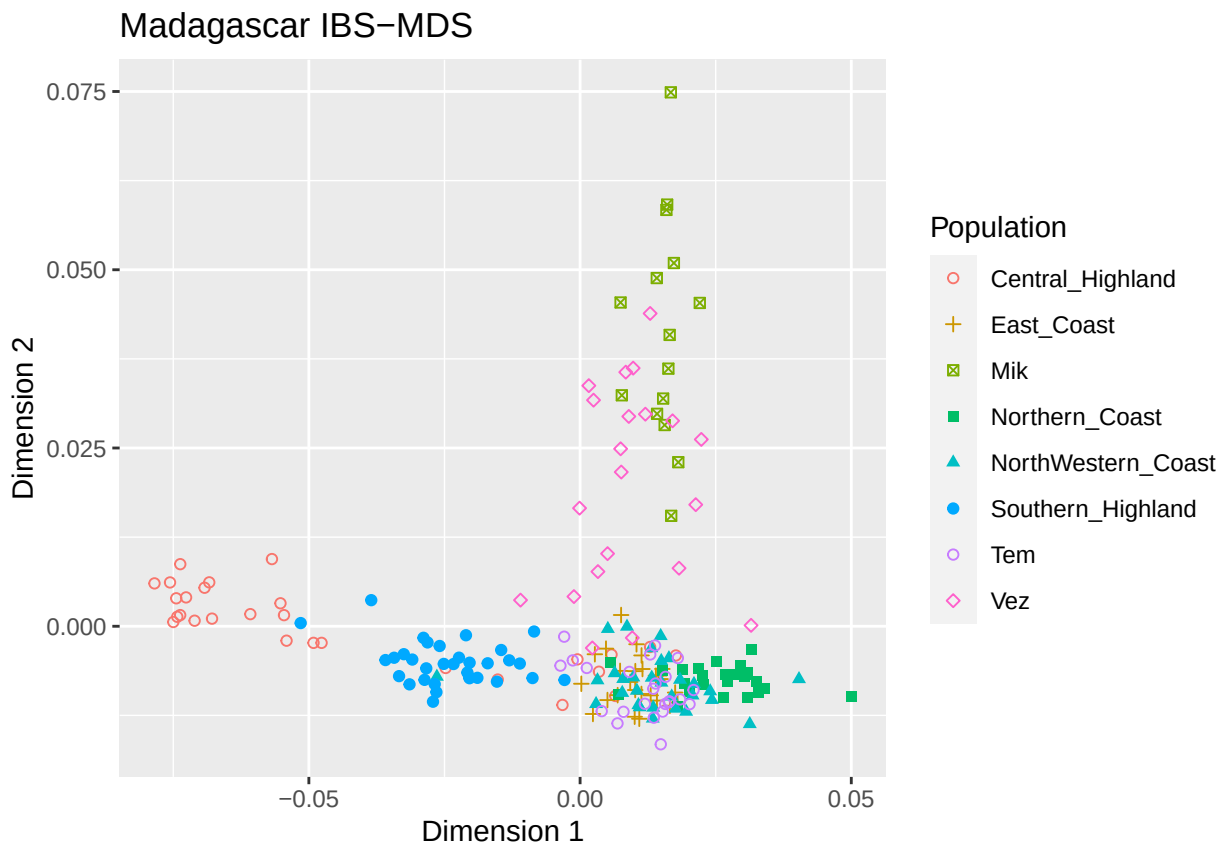
```
## [1] 0.12755136 0.08220343 0.07715788
```

```
## [1] 0.016269350 0.006757405 0.005953339
```

```
## [1] 0.02898009
```

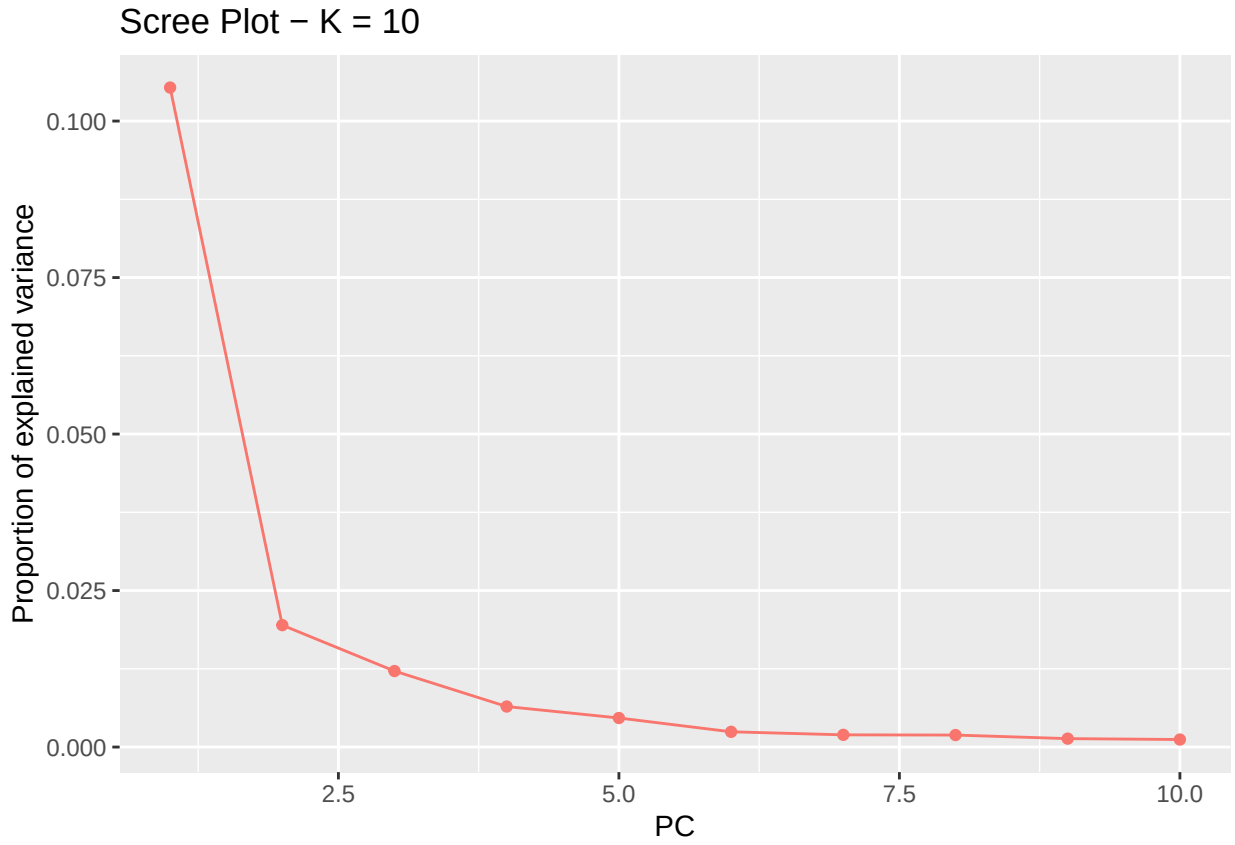


PCA of Malagasy populations alone shows overlap in PC space of the Betsimisaraka, Sakalava, Tsimhety, and Diego samples, all from Northern Madagascar. Some Merina samples also fall within this cluster, but the Merina and Betsileo (more Southern) populations show less overlap with others. However, the overall % explained by each PC is very low.

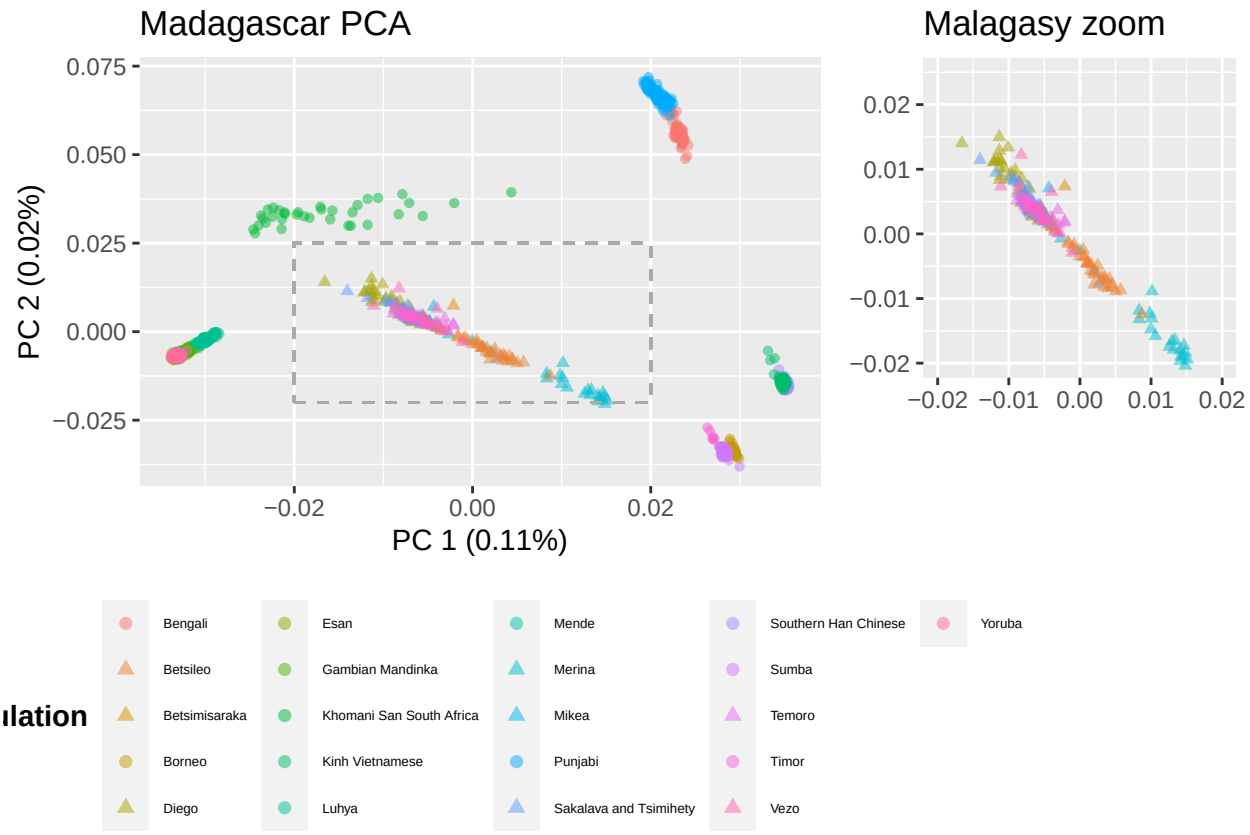


IBS-MDS exhibits an almost identical patterning of popualtions to the PCA above.

PCA's of our merged data (so far)

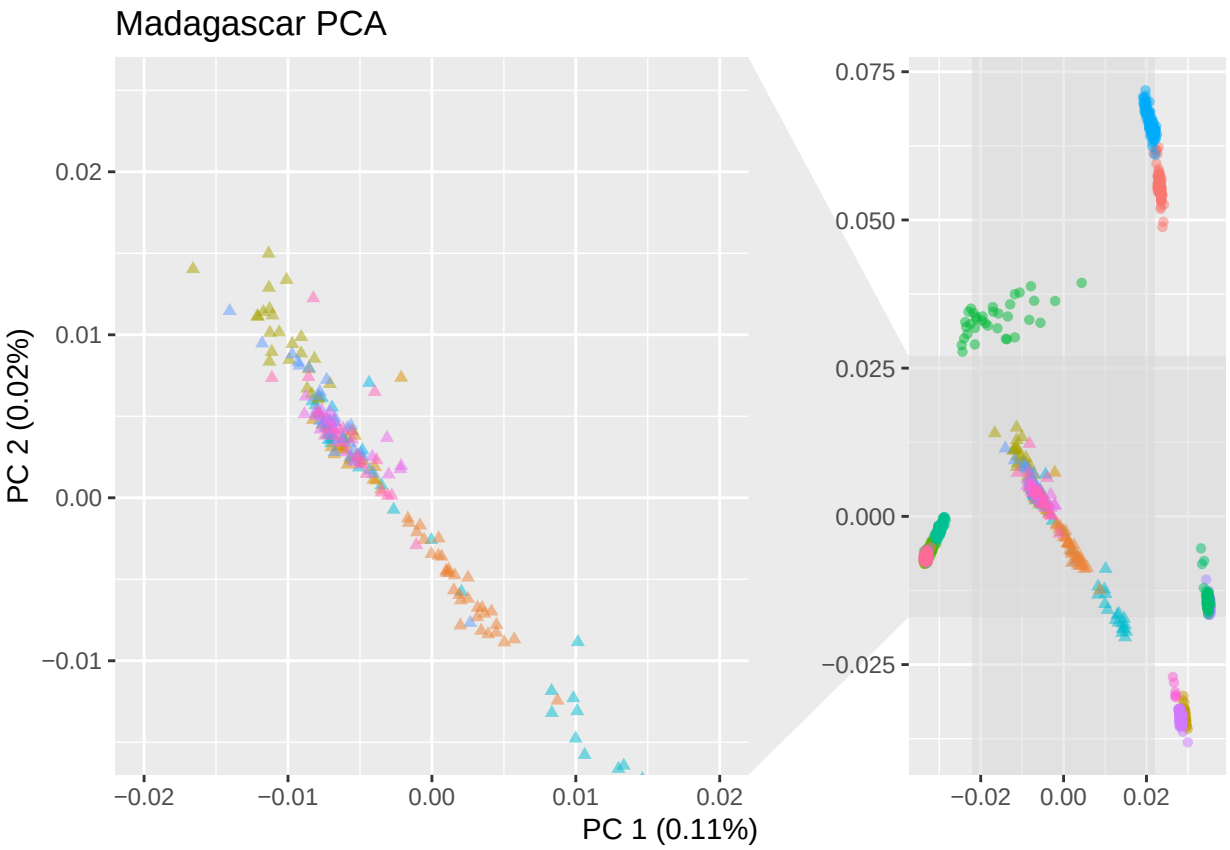



```
## [1] 0.32458210 0.13954654 0.11019244 0.08045750 0.06814429
## [1] 0.105353538 0.019473238 0.012142374 0.006473410 0.004643644
## [1] 0.1480862
```



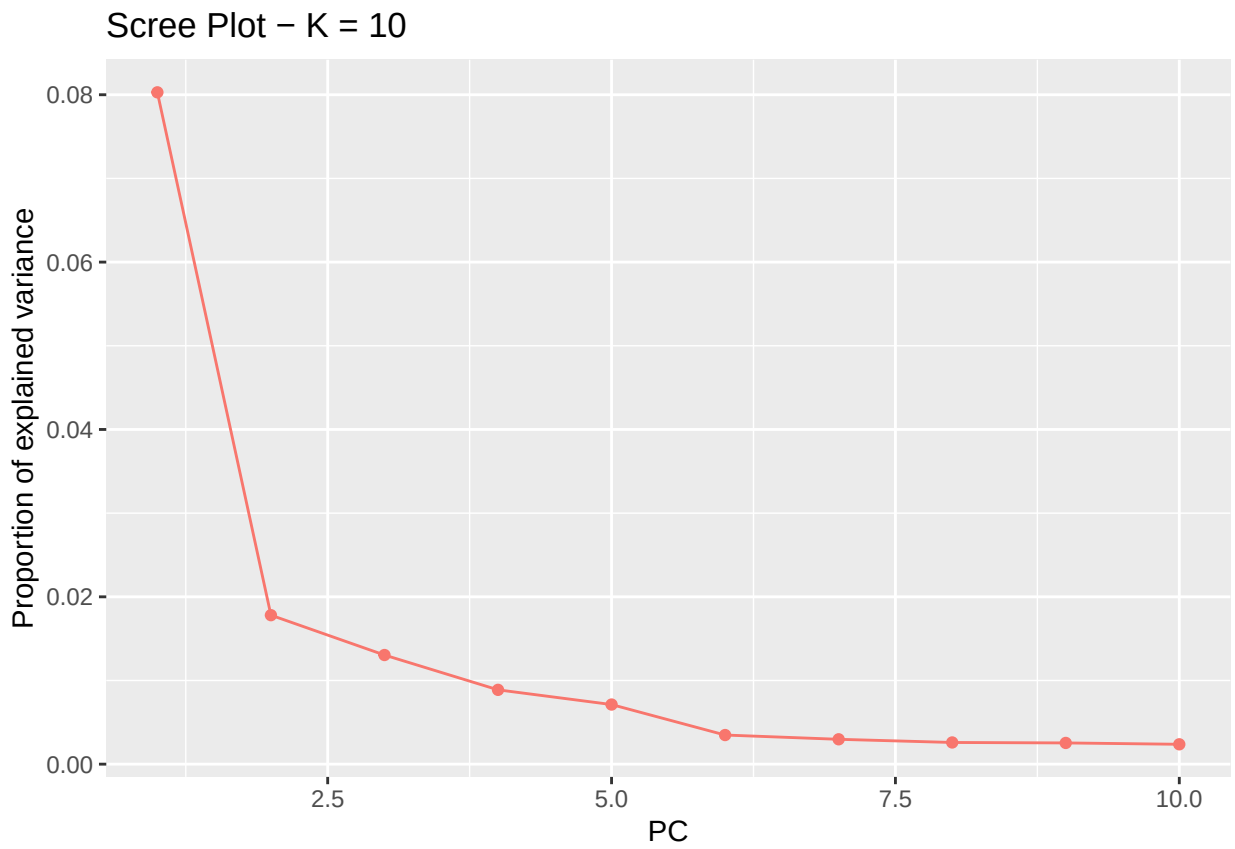
In the first two PC's, the Malagasy form a cline between African and southeast Asian groups. The Merina are located closer to individuals from Borneo and the Lesser Sunda islands while Northern Malagasy groups are located towards Southern African and Bantu individuals.

Facet zoom inset plot version

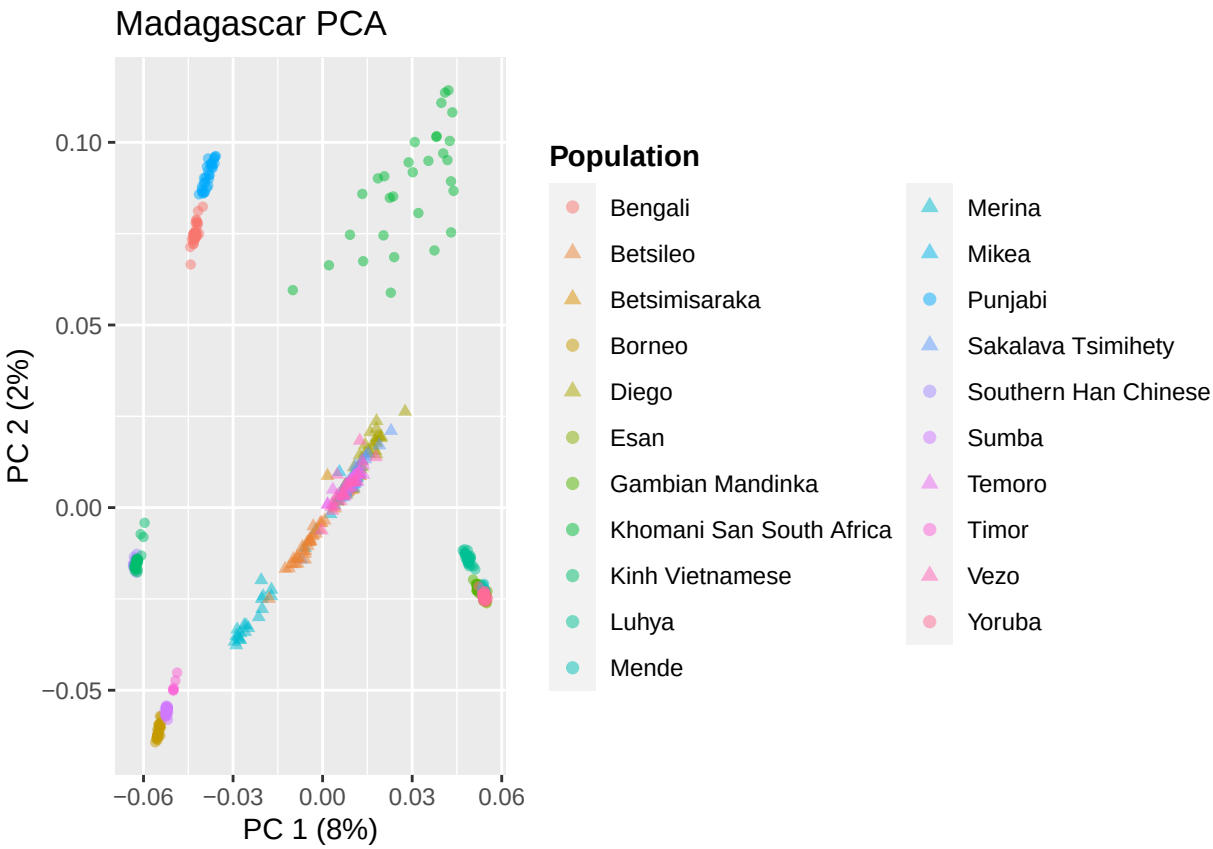


PCA's using a subsample of individuals

Here, individuals are sampled from each population to even out sample size bias. 30 individuals are sampled from each population, if populations have less than 30 individuals all are included.

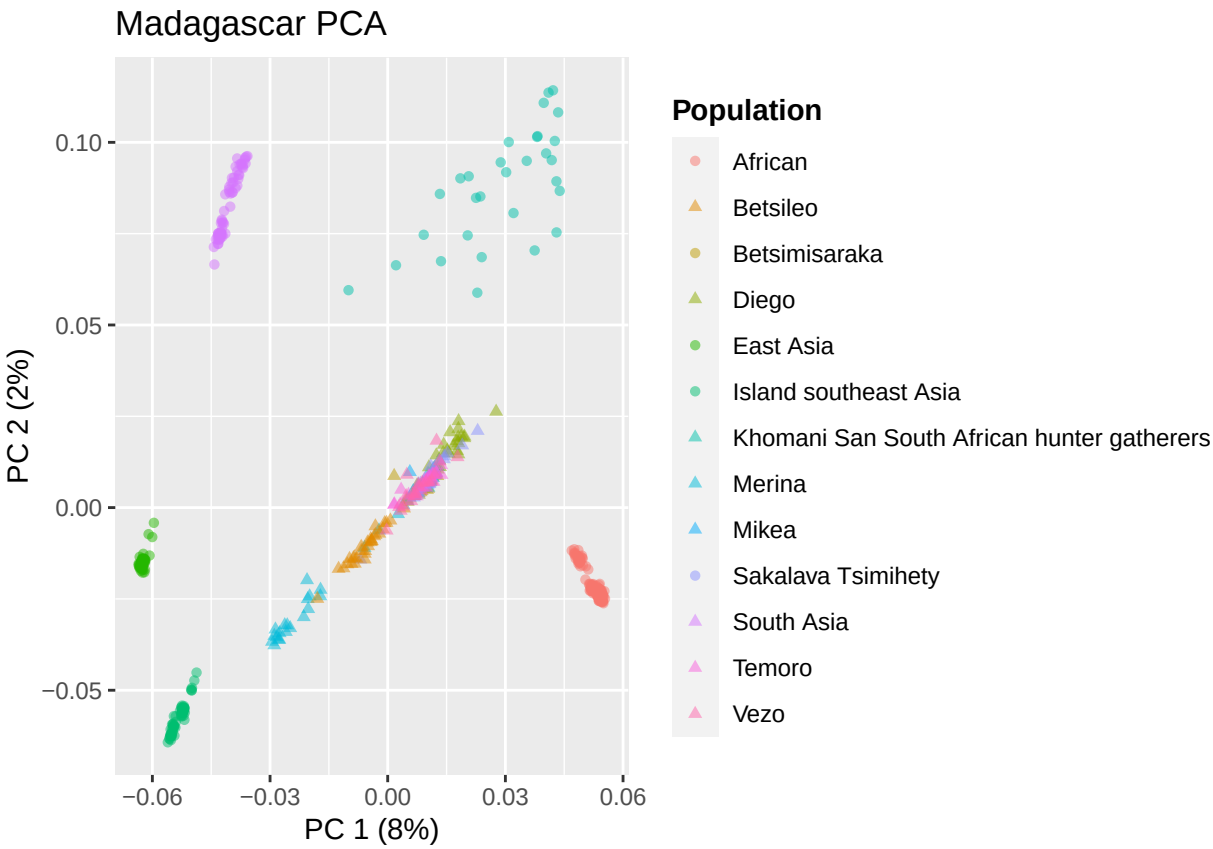



```
## [1] 0.28335359 0.13343816 0.11420784 0.09424448
## [1] 0.080289256 0.017805743 0.013043431 0.008882023
## [1] 0.1200205
```



Subsampled PCA with broader groupings

Plot using broad continental sample groupings instead of population ID's.


Madagascar PCA
PC 2 (2%)
PC 1 (8%)
0.10
0.05
0.00
−0.05
−0.06
−0.03
0.00
0.03
0.06
Population
African
Betsileo
Betsimisaraka
Diego
East Asia
Island southeast Asia
Khomani San South African hunter gatherers
Merina
Mikea
Sakalava Tsimihety
South Asia
Temoro
Vezo